

Laboratory Equipment and Safety

Hazards, glassware and reading scales

Name _____ Class _____ Date _____

TYPICAL TOLERANCES (CLASS A)

100 mL measuring cylinder ± 0.5 mL · 10 mL pipette ± 0.02 mL · 25 mL pipette ± 0.03 mL

50 mL burette: each reading ± 0.05 mL · 250 mL volumetric flask ± 0.15 mL

Top-pan balance ± 0.01 g · analytical balance ± 0.0001 g

Percent uncertainty = $\text{uncertainty} \div \text{measured value} \times 100\%$. Show units in every step.

A. Hazards

1. State what each GHS pictogram means and give one example of a substance that carries it:
(a) flame (b) flame over circle (c) corrosion (d) skull and crossbones (e) environment

2. (a) Which signal word shows the more serious hazard? (b) What is a safety data sheet, and when should you read it?

3. Identify **five** unsafe actions in this account and say what should be done instead: “Sam pushed her goggles up onto her head to see better, poured water into concentrated sulfuric acid, then heated ethanol in a test tube over a Bunsen flame, pointing the tube at her partner. She sipped her drink while waiting and poured the leftover copper(II) sulfate solution down the sink.”

B. Choosing equipment

4. Choose the best equipment for each task: (a) adding exactly 25.00 mL of a solution to a flask (b) roughly 50 mL of water to dissolve a solid (c) making exactly 250.0 mL of a solution (d) adding acid drop by drop while measuring the volume (e) weighing 0.5000 g of a solid

5. Calculate the percent uncertainty of: (a) 0.50 g on a top-pan balance and 0.5000 g on an analytical balance (b) 10.0 mL measured with a 100 mL measuring cylinder and 10.00 mL delivered by a 10 mL pipette. What do your answers show?

C. Burette readings

6. A student records these titrations. Rough: 0.00 mL $\rightarrow$ 24.10 mL. Run 1: 1.20 mL $\rightarrow$ 24.70 mL. Run 2: 0.40 mL $\rightarrow$ 23.85 mL. Run 3: 0.05 mL $\rightarrow$ 23.55 mL. (a) Calculate each titre. (b) Calculate the mean and standard deviation of runs 1–3 and report the result.

7. Explain why you read the bottom of the meniscus with your eye level with it, and name the error caused by reading from above.

D. Challenge

8. Each burette reading is uncertain by ± 0.05 mL, and a titre is the difference of two readings. (a) Calculate the percent uncertainty of a titre of 23.50 mL. (b) Suggest one way to reduce the percent uncertainty of the titre.